

## Week 3

### Problem 1 -Number of days between two days

```
import java.time.LocalDate;
import java.time.temporal.ChronoUnit;
class Solution {
    public int daysBetweenDates(String date1, String date2) {
        LocalDate d1 = LocalDate.parse(date1);
        LocalDate d2 = LocalDate.parse(date2);
        return (int) Math.abs(ChronoUnit.DAYS.between(d1, d2));
    }
}
public class Main {
    public static void main(String[] args) {
        Solution solver = new Solution();
        String test1_date1 = "2019-06-29";
        String test1_date2 = "2019-06-30";
        int result1 = solver.daysBetweenDates(test1_date1, test1_date2);
        System.out.println("Example 1 Output: " + result1);
        String test2_date1 = "2020-01-15";
        String test2_date2 = "2019-12-31";
        int result2 = solver.daysBetweenDates(test2_date1, test2_date2);
        System.out.println("Example 2 Output: " + result2);
    }
}
```

## Output

Input 1

date1 =

"2019-06-29"

date2 =

"2019-06-30"

Output

1

Input 2

date1 =

"2020-01-15"

date2 =

"2019-12-31"

Output

15

## Problem 2-Day of the Week

```
import java.time.LocalDate;
```

```
import java.time.format.TextStyle;
```

```
import java.util.Locale;
```

```
public class Solution {
```

```
    public String dayOfTheWeek(int day, int month, int year) {
```

```
        LocalDate date = LocalDate.of(year, month, day);
```

```
        return date.getDayOfWeek().getDisplayName(TextStyle.FULL, Locale.ENGLISH);
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        Solution solution = new Solution();
```

```
        // Test Case 1
```

```
        int day1 = 31, month1 = 8, year1 = 2019;
```

```
        System.out.println("Input: day = " + day1 + ", month = " + month1 + ", year = " + year1);
```

```
        System.out.println("Output: \"" + solution.dayOfTheWeek(day1, month1, year1) + "\"");
```

```
        System.out.println();
```

```
        // Test Case 2
```

```
        int day2 = 18, month2 = 7, year2 = 1999;
```

```
        System.out.println("Input: day = " + day2 + ", month = " + month2 + ", year = " + year2);
```

```
        System.out.println("Output: \"" + solution.dayOfTheWeek(day2, month2, year2) + "\"");
```

```
        System.out.println();
```

```
        // Test Case 3
```

```
        int day3 = 15, month3 = 8, year3 = 1993;
```

```
        System.out.println("Input: day = " + day3 + ", month = " + month3 + ", year = " + year3);
        System.out.println("Output: \"" + solution.dayOfTheWeek(day3, month3, year3) + "\"");
    }
}
```

## Output-

Input 1

day =

31

month =

8

year =

2019

Output

"Saturday"

Input 2

day =

18

month =

7

year =

1999

Output

"Sunday"

Input 3

day =

15

month =

8

year =

1993

Output

"Sunday"

## Problem 3-Day of the Year

```
import java.time.LocalDate;

public class Solution {

    public int dayOfYear(String date) {
        // Parse the YYYY-MM-DD string into a LocalDate object
        LocalDate localDate = LocalDate.parse(date);
        return localDate.getDayOfYear();
    }

    public static void main(String[] args) {
        Solution solution = new Solution();

        // Test Case 1
        String date1 = "2019-01-09";
        System.out.println("Input: date = " + date1);
        int result1 = solution.dayOfYear(date1);
        System.out.println("Output: " + result1); // Expected: 9
        System.out.println();

        // Test Case 2
        String date2 = "2019-02-10";
        System.out.println("Input: date = " + date2);
        int result2 = solution.dayOfYear(date2);
        System.out.println("Output: " + result2); // Expected: 41
        System.out.println();

        // Test Case 3 (Leap Year Example)
        String date3 = "2012-03-01";
        System.out.println("Input: date = " + date3);
        int result3 = solution.dayOfYear(date3);
        System.out.println("Output: " + result3); // Expected: 61 (Jan=31 + Feb=29 + Mar=1)
    }
}
```

Output-

Input 1

date =

"2019-01-09"

Output

9

Input 2

date =

"2019-02-10"

Output

41

## Problem 4 -Java array list

```
import java.io.*;
import java.util.*;

public class Solution {

    public static void main(String[] args) {

        /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class
        should be named Solution. */

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        ArrayList<ArrayList<Integer>> list = new ArrayList<>();

        for (int i = 0; i < n; i++) {

            int d = sc.nextInt();

            ArrayList<Integer> row = new ArrayList<>();

            for (int j = 0; j < d; j++) {

                row.add(sc.nextInt());

            }

            list.add(row);

        }

        int q = sc.nextInt();

        for (int i = 0; i < q; i++) {
```

```

        int x = sc.nextInt();
        int y = sc.nextInt();
        if (x <= list.size() && y <= list.get(x - 1).size()) {
            System.out.println(list.get(x - 1).get(y - 1));
        } else {
            System.out.println("ERROR!");
        }
    }
}
sc.close();
}
}

```

## Output-

Compiler Message

Success

Input (stdin)

5

5 41 77 74 22 44

1 12

4 37 34 36 52

0

3 20 22 33

5

1 3

3 4

3 1

4 3

5 5

Expected Output

74

52

37

ERROR!

ERROR!

## Problem 5-Java date and time

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;

class Result {

    public static String findDay(int month, int day, int year) {

        Calendar cal = Calendar.getInstance();
        cal.set(year, month - 1, day);

        return new SimpleDateFormat("EEEE", Locale.ENGLISH)
            .format(cal.getTime())
            .toUpperCase();
    }
}

public class Solution {

    public static void main(String[] args) throws IOException {

        BufferedReader bufferedReader = new BufferedReader(new
        InputStreamReader(System.in));

        BufferedWriter bufferedWriter = new BufferedWriter(new
        FileWriter(System.getenv("OUTPUT_PATH")));

        String[] firstMultipleInput = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

        int month = Integer.parseInt(firstMultipleInput[0]);
```

```

        int day = Integer.parseInt(firstMultipleInput[1]);
        int year = Integer.parseInt(firstMultipleInput[2])
        String res = Result.findDay(month, day, year)
        bufferedWriter.write(res);
        bufferedWriter.newLine();
        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

## Output-

Compiler Message

Success

Input (stdin)

08 05 2015

Expected Output

WEDNESDAY

## Problem 6-Java priority queue

```

import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;
import java.util.PriorityQueue;
class Student {
    private int id;
    private String name;
    private double cgpa;
    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }
    public int getID() {

```



```

        return id;
    }
    public String getName() {
        return name;
    }
    public double getCGPA() {
        return cgpa;
    }
}

class Priorities {
    public List<Student> getStudents(List<String> events) {
        PriorityQueue<Student> pq = new PriorityQueue<>(
            (a, b) -> {
                int result = Double.compare(b.getCGPA(), a.getCGPA());

                if (result != 0) {
                    return result;
                }
                result = a.getName().compareTo(b.getName());
                if (result != 0) {
                    return result;
                }
                return Integer.compare(a.getID(), b.getID());
            }
        );
        for (String event : events) {
            String[] data = event.split(" ");
            if (data[0].equals("ENTER")) {
                String name = data[1];
                double cgpa = Double.parseDouble(data[2]);
                int id = Integer.parseInt(data[3]);
                pq.add(new Student(id, name, cgpa));
            }
        }
    }
}

```

```

        } else if (data[0].equals("SERVED")) {
            if (!pq.isEmpty()) {
                pq.poll();
            }
        }
    }

    List<Student> result = new ArrayList<>();
    while (!pq.isEmpty()) {
        result.add(pq.poll());
    }
    return result;
}

}

public class Solution {
    private final static Scanner scan = new Scanner(System.in);
    private final static Priorities priorities = new Priorities();

    public static void main(String[] args) {
        int totalEvents = Integer.parseInt(scan.nextLine());
        List<String> events = new ArrayList<>();
        while (totalEvents-- != 0) {
            String event = scan.nextLine();
            events.add(event);
        }

        List<Student> students = priorities.getStudents(events);
        if (students.isEmpty()) {
            System.out.println("EMPTY");
        } else {
            for (Student st: students) {
                System.out.println(st.getName());
            }
        }
    }
}

```

```
}
```

## Output –

Compiler Message

Success

Input (stdin)

12

ENTER John 3.75 50

ENTER Mark 3.8 24

ENTER Shafaet 3.7 35

SERVED

SERVED

ENTER Samiha 3.85 36

SERVED

ENTER Ashley 3.9 42

ENTER Maria 3.6 46

ENTER Anik 3.95 49

ENTER Dan 3.95 50

SERVED

Expected Output

Dan

Ashley

Shafaet

Maria

## Problem 7-Sort characters by frequency

```
class Solution {  
    public String frequencySort(String s) {  
        int[] freq = new int[128];  
  
        for (char c : s.toCharArray()) {  
            freq[c]++;  
        }  
    }  
}
```

```

StringBuilder[] buckets = new StringBuilder[s.length() + 1];

for (int i = 0; i < freq.length; i++) {
    if (freq[i] > 0) {
        if (buckets[freq[i]] == null) {
            buckets[freq[i]] = new StringBuilder();
        }
        buckets[freq[i]].append((char) i);
    }
}

StringBuilder result = new StringBuilder();
for (int count = s.length(); count >= 1; count--) {
    if (buckets[count] == null) {
        continue;
    }
    for (char c : buckets[count].toString().toCharArray()) {
        result.append(c);
    }
}

return result.toString();
}
}

```

## Output-

Input 1

s = "tree"

output

"eert"

Input 2

s = "cccaa"

Output

"aaaccc"

Input 3

s = "Aabb"

output

"bbAa"

## Problem 8- Largest Number

```
import java.util.Arrays;
```

```
import java.util.Comparator;
```

```
public class Solution {
```

```
    public String largestNumber(int[] nums) {
```

```
        String[] asStrs = new String[nums.length];
```

```
        for (int i = 0; i < nums.length; i++) {
```

```
            asStrs[i] = String.valueOf(nums[i]);
```

```
        }
```

```
        Arrays.sort(asStrs, new Comparator<String>() {
```

```
            @Override
```

```
            public int compare(String s1, String s2) {
```

```
                String order1 = s1 + s2;
```

```
                String order2 = s2 + s1;
```

```
                return order2.compareTo(order1);
```

```
            }
```

```
        });
```

```
        if (asStrs[0].equals("0")) {
```

```
            return "0";
```

```
        }
```

```
        StringBuilder largestNumberStr = new StringBuilder();
```

```

        for (String numAsStr : asStrs) {
            largestNumberStr.append(numAsStr);
        }

        return largestNumberStr.toString();
    }

    public static void main(String[] args) {
        Solution solution = new Solution();
        int[] nums1 = {10, 2};
        System.out.println("Input: " + Arrays.toString(nums1));
        System.out.println("Output: \"" + solution.largestNumber(nums1) + "\"");
        System.out.println();
        int[] nums2 = {3, 30, 34, 5, 9};
        System.out.println("Input: " + Arrays.toString(nums2));
        System.out.println("Output: \"" + solution.largestNumber(nums2) + "\"");
        System.out.println();
        int[] nums3 = {0, 0, 0};
        System.out.println("Input: " + Arrays.toString(nums3));
        System.out.println("Output: \"" + solution.largestNumber(nums3) + "\"");
    }
}

```

## Output-

Input 1

nums =

[10,2]

Output

"210"

Input 2

nums =

[3,30,34,5,9]

Output

"9534330"

# Problem 9-Java Sort

```
import java.util.*;

class Student{
    private int id;
    private String fname;
    private double cgpa;
    public Student(int id, String fname, double cgpa) {
        super();
        this.id = id;
        this.fname = fname;
        this.cgpa = cgpa;
    }
    public int getId() {
        return id;
    }
    public String getFname() {
        return fname;
    }
    public double getCgpa() {
        return cgpa;
    }
}

public class Solution
{
    public static void main(String[] args){
        Scanner in = new Scanner(System.in);
        int testCases = Integer.parseInt(in.nextLine());

        List<Student> studentList = new ArrayList<Student>();
        while(testCases>0){
            int id = in.nextInt();
            String fname = in.next();
```

```

        double cgpa = in.nextDouble();

        Student st = new Student(id, fname, cgpa);
        studentList.add(st);
        testCases--;
    }

    Collections.sort(studentList, new Comparator<Student>() {
        @Override
        public int compare(Student s1, Student s2) {
            if (s1.getCgpa() != s2.getCgpa()) {
                return Double.compare(s2.getCgpa(), s1.getCgpa());
            }
            int nameCompare =
                s1.getFname().compareTo(s2.getFname());
            if (nameCompare != 0) {
                return nameCompare;
            }
            return Integer.compare(s1.getId(), s2.getId());
        }
    });

    for(Student st: studentList){
        System.out.println(st.getFname());
    }
}

```

## Output-

Compiler Message

Success

Input (stdin)

5

33 Rumpa 3.68

85 Ashis 3.85

56 Samiha 3.75



19 Samara 3.75

22 Fahim 3.76

Expected Output

Ashis

Fahim

Samara

Samiha

Rumpa

## Problem 10 – Java Comparator

```
import java.util.*;

class Checker implements Comparator<Player> {

    @Override
    public int compare(Player p1, Player p2) {
        if (p1.score != p2.score) {
            return p2.score - p1.score;
        }
        return p1.name.compareTo(p2.name);
    }
}

class Player{
    String name;
    int score;

    Player(String name, int score){
        this.name = name;
        this.score = score;
    }
}

class Solution {

    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        int n = scan.nextInt();
        Player[] player = new Player[n];
```

```

        Checker checker = new Checker();
        for(int i = 0; i < n; i++){
            player[i] = new Player(scan.next(), scan.nextInt());
        }
        scan.close();
        Arrays.sort(player, checker);
        for(int i = 0; i < player.length; i++){
            System.out.printf("%s %s\n", player[i].name, player[i].score);
        }
    }
}

```

## Output-

Input (stdin)

5

amy 100

david 100

heraldo 50

aakansha 75

aleksa 150

Your Output (stdout)

aleksa 150

amy 100

david 100

aakansha 75

heraldo 50

Expected Output

aleksa 150

amy 100

david 100

aakansha 75

heraldo 50

